

Galaxy for virologist training. Exercise 2: Quality control and trimming

Despite the improvement of sequencing methods, there is no error-free technique. A correct measuring of the sequencing quality is essential to identify issues in the sequencing process; thus, this must be the first step in every sequencing analysis. Once the quality control is finished, it's important to remove low quality reads, or short reads, which is why a subsequent trimming step is mandatory. After the trimming step, it is recommended to perform a new quality control step to be sure that trimming worked correctly.

1. Illumina Quality control and trimming

Title	Pre-processing
Training dataset:	PRJEB43037 - In August 2020, an outbreak of West Nile Virus affected 71 people with meningoencephalitis in Andalusia and 6 more people in Extremadura (south-west of Spain), causing a total of eight deaths. The virus belonged to the lineage 1 and was relatively similar to previous outbreaks occurred in the Mediterranean region. Here, we present a detailed analysis of the outbreak, including an extensive phylogenetic study. This is one of the outbreak samples.
Questions:	• How do I check whether my Illumina data was correctly sequenced? • How can I improve the quality of my data?
Objectives:	• Perform a quality control in raw Illumina reads. • Perform a quality trimming in raw Illumina reads. • Perform a quality control in trimmed Illumina reads.
Estimated time:	25 min

1.1. Quality control

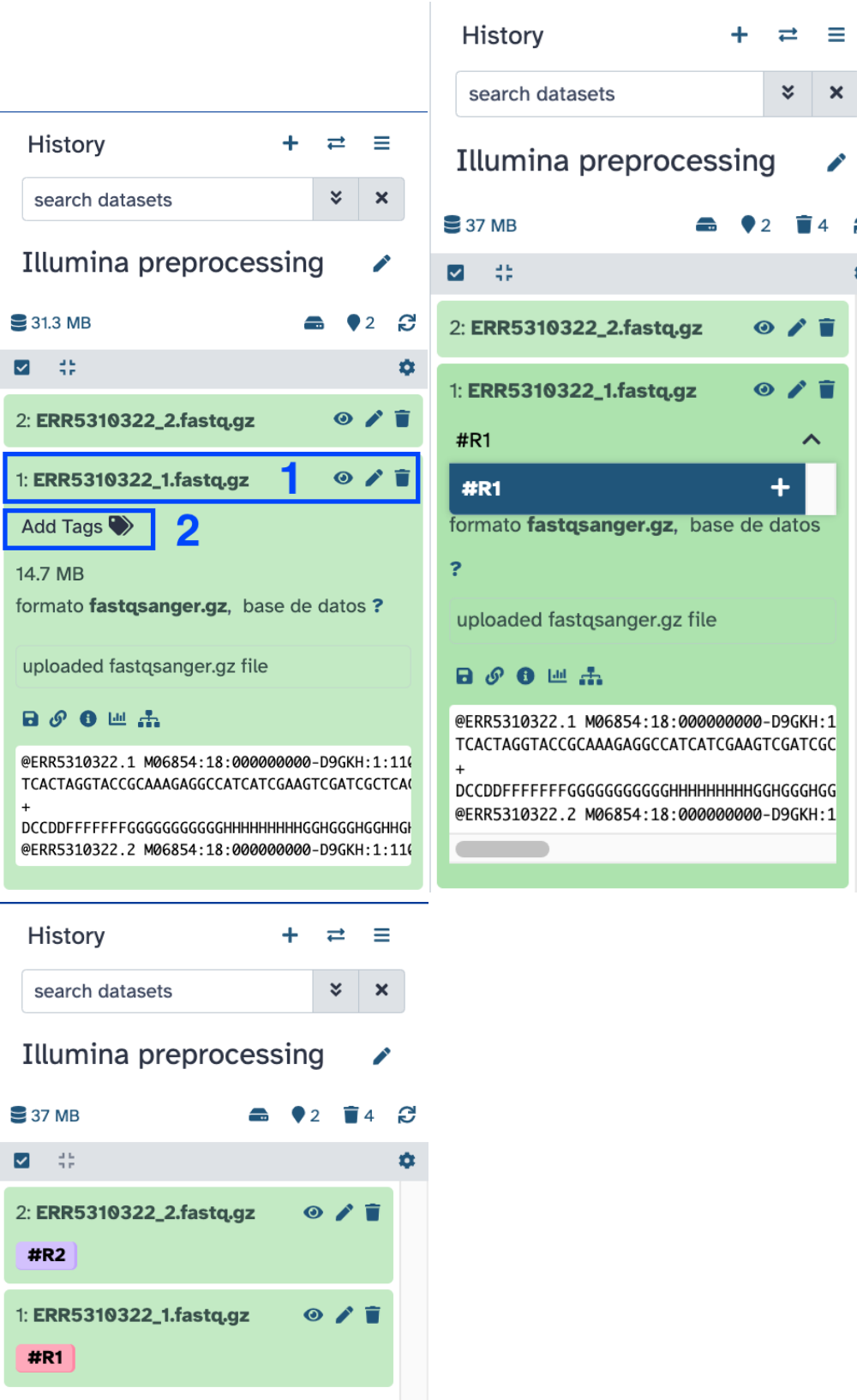
1.1.1. Upload data

To run the quality control over the samples, follow these steps:

1. [Create a new history, as we explained yesterday](#) named **Illumina preprocessing**.
2. [Upload data as seen yesterday](#), copy and paste the following URLs:

```
ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR531/002/ERR5310322/ERR5310322_1.fastq.gz
ftp://ftp.sra.ebi.ac.uk/vol1/fastq/ERR531/002/ERR5310322/ERR5310322_2.fastq.gz
```

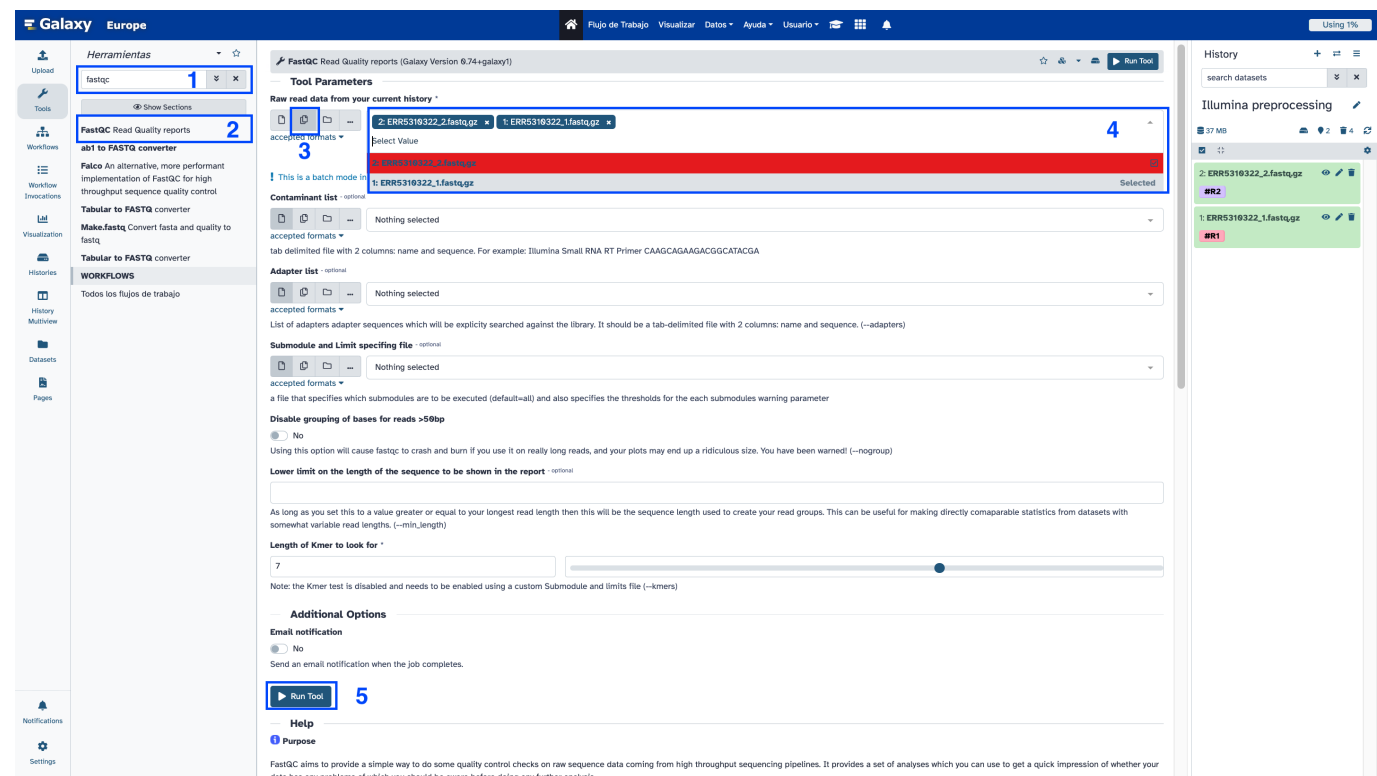
- 3. Add some tags to the files. *It is mandatory that the tag starts with # to be propagated to the processes.*



1.1.2. Run FastQC

- 1. Search for the **FastQC** tool.
- 2. Select **FastQC Read Quality reports** and then set the following parameters:
- 3. Select multiple file data set in Raw read data from your current history.
- 4. Select the two datasets.

5. Then go down and select **Run tool**.



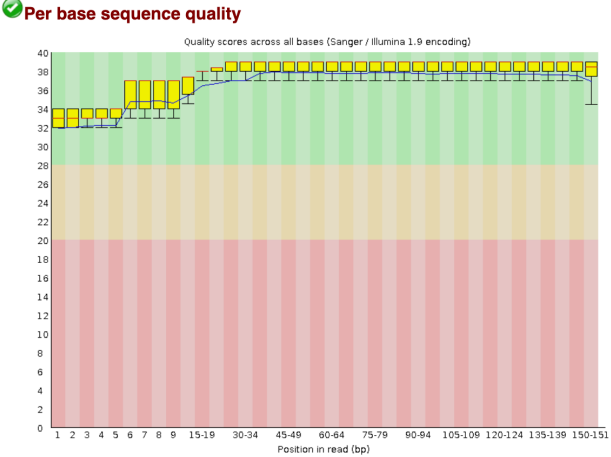
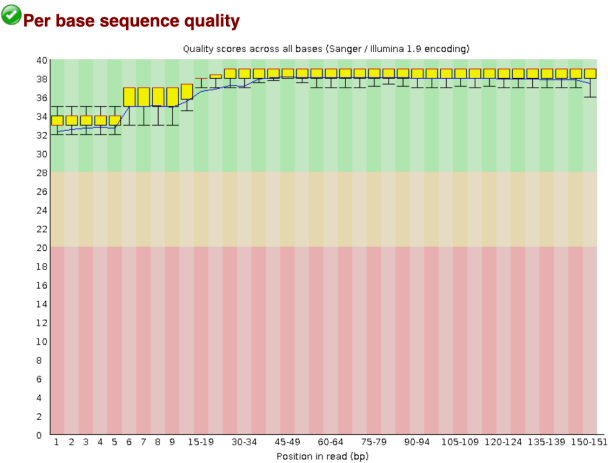
To see the results we are going to open the jobs with **Web page** in their name for both data 1 and data 2.

✔ **Basic Statistics**

Measure	Value
Filename	ERR5310322_1_fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	265989
Sequences flagged as poor quality	0
Sequence length	35-151
%GC	51

✔ **Basic Statistics**

Measure	Value
Filename	ERR5310322_2_fastq.gz
File type	Conventional base calls
Encoding	Sanger / Illumina 1.9
Total Sequences	265989
Sequences flagged as poor quality	0
Sequence length	35-151
%GC	51



Here, you can see the number of reads in each file, the maximum and minimum length of all reads in the sample, and the quality plots for both R1 and R2. They look quite good, but we are going to run trimming over the samples.

- ▶ How many reads do the samples have?
- ▶ How do I check whether my Illumina data was correctly sequenced?

1.2. Trimming

Once we have performed the quality control, we have to perform the quality and read length trimming:

1.2.1. Run Fastp

1. Search for **fastp** in the tools.
2. Then select **fastp - fast all-in-one preprocessing for FASTQ files**.
3. Single-end or paired reads > **Paired Collection**

[!WARNING] To use this option from Fastp, it is necessary to first create a **list of paired datasets**, given our .fastq files, which will then be used for the rest of the steps of this workshop. In order to do so, we must:

1. Select our fastqsanger.gz files that we uploaded in the beginning, and then click on *2 of X selected*.
2. Select *Advanced Build List*.
3. Select the *List of Paired Datasets* box and click on *Next*.
4. Check the auto-pairing configuration and click again on *Next*.
5. Give a name to your list and finally click on *Build*.

Once this is done, you'll be able to select your list as input of Fastp.

4. Display **Filter Options**

- Quality Filtering options
 - Qualified Quality Phred = **30**
 - Unqualified percent limit = **10**
- Length Filtering Options
 - Length required = **50**

5. Read modification options

- PoliX tail trimming > **Enable polyX tail trimming**
- Per read cutting by quality options
 - Cut by quality in front (5') > **Yes**
 - Cut by quality in tail (3') > **Yes**
 - Cutting mean quality = **30**

6. Finally, click on **Run tool**.

Upload

Tools

Workflows

Workflow Invocations

Visualization

Histories

History Multiview

Datasets

Pages

Notifications

Settings

Herramientas

fastp

1

Show Sections

fastp last all-in-one preprocessing for FASTQ files

2

fastpca - dimensionality reduction of MD simulations

Map with minimap2 A fast pairwise aligner for genomic and spliced nucleotide sequences

Chromester ultra-fast pairwise genome comparisons

RNAJOB Fast Pattern searching for RNA secondary structures

fgsea - fast preranked gene set enrichment analysis

WORKFLOWS

Todos los flujos de trabajo

fastp last all-in-one preprocessing for FASTQ files (Galaxy Version 9.23.4+galaxy2)

Flujo de Trabajo Visualizar Datos Ayuda Usuario

Run Tool

Tool Parameters

Single-end or paired reads

Paired

3

Input 1

4

accepted formats

Input FASTQ file #1 (-i)

Input 2

5

accepted formats

Input FASTQ file #2 (-I)

Merge forward and reverse reads?

No

(--merge)

Adapter Trimming Options

Global trimming options

Overrepresented Sequence Analysis

Filter Options

6

Quality filtering options

Disable quality filtering

No

Quality filtering is enabled by default. If this option is specified, quality filtering is disabled. (-q)

Qualified quality phred

30

The quality value that a base is qualified. Default: 15 means phred quality >=Q15 is qualified. (-q)

Unqualified percent limit

10

How many percents of bases are allowed to be unqualified (0-100). Default 40 means 40%. (-u)

N base limit

If one read's number of N base is >n_base_limit, then this read/pair is discarded. Default is 5. (-n)

History

search datasets

Ilumina preprocessing

42.7 MB

10: FastQC on data 2: RawData

9: FastQC on data 2: Webpage

8: FastQC on data 1: RawData

7: FastQC on data 1: Webpage

2: ERRS310322_2.fastq.gz

1: ERRS310322_1.fastq.gz

Upload

Tools

Workflows

Workflow Invocations

Visualization

Histories

History Multiview

Datasets

Pages

Notifications

Settings

Herramientas

fastp

1

Show Sections

fastp last all-in-one preprocessing for FASTQ files

2

fastpca - dimensionality reduction of MD simulations

Map with minimap2 A fast pairwise aligner for genomic and spliced nucleotide sequences

Chromester ultra-fast pairwise genome comparisons

RNAJOB Fast Pattern searching for RNA secondary structures

fgsea - fast preranked gene set enrichment analysis

WORKFLOWS

Todos los flujos de trabajo

fastp last all-in-one preprocessing for FASTQ files (Galaxy Version 9.23.4+galaxy2)

Flujo de Trabajo Visualizar Datos Ayuda Usuario

Run Tool

Tool Parameters

Length filtering options

Disable length filtering

No

Length filtering is enabled by default. If this option is specified, length filtering is disabled. (-l)

Length required

50

9

Reads shorter than this value will be discarded. Default is 15. (-l)

Maximum length

Reads longer than this value will be discarded. Default is 0 and means no limitation. (--length_limit)

Low complexity filtering options

Enable low complexity filter

No

The complexity is defined as the percentage of base that is different from its next base, default is No (-y)

Complexity threshold

Threshold for low complexity filter (0-100). Default is 30, which means 30% complexity is required. (-Y)

Read Modification Options

10

PolyG tail trimming

Automatic trimming for Illumina NextSeq/Novaseq data

This feature is enabled for NextSeq/Novaseq data by default. NextSeq/Novaseq data is detected by the machine ID in the FASTQ records.

PolyG minimum length

The minimum length to detect polyG in the read tail. 10 by default. (--poly_g_min_len)

PolyX tail trimming

Enable polyX tail trimming

11

Similar to polyG tail trimming. When polyG tail trimming and polyX tail trimming are both enabled, fastp will perform polyG trimming first, then perform polyX trimming. Disabled by default.

PolyX minimum length

The minimum length to detect polyX in the read tail. 10 by default. (--poly_x_min_len)

UMI processing

History

search datasets

Ilumina preprocessing

42.7 MB

10: FastQC on data 2: RawData

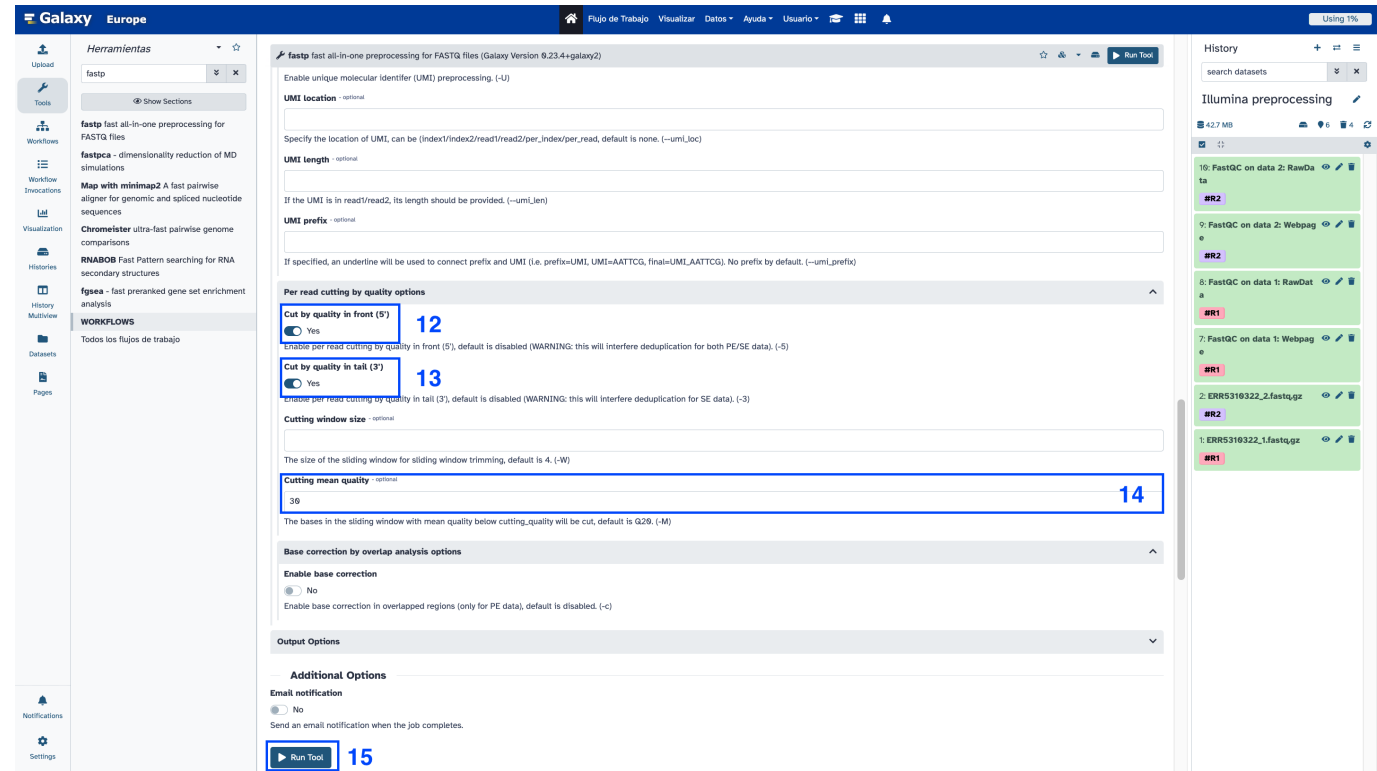
9: FastQC on data 2: Webpage

8: FastQC on data 1: RawData

7: FastQC on data 1: Webpage

2: ERRS310322_2.fastq.gz

1: ERRS310322_1.fastq.gz



To see the trimming stats, have a look at the **fastp on collection X: HTML report** file. You should see something like this.

fastp report for ERR5310322_1_fastq.gz.fastq.gz

Summary

General

fastp version:	0.20.1 (https://github.com/OpenGene/fastp)
sequencing:	paired end (151 cycles + 151 cycles)
mean length before filtering:	105bp, 105bp
mean length after filtering:	113bp, 113bp
duplication rate:	19.977989%
Insert size peak:	84

Before filtering

total reads:	531.978000 K
total bases:	56.257825 M
Q20 bases:	54.842431 M (97.484094%)
Q30 bases:	54.605191 M (97.062393%)
GC content:	50.644494%

After filtering

total reads:	433.314000 K
total bases:	49.003611 M
Q20 bases:	48.876432 M (99.740470%)
Q30 bases:	48.825481 M (99.636496%)
GC content:	51.087943%

Filtering result

- How many reads have we lost?

1.2.2. Other trimming tools: Trimmomatic

1. Search for **trimmomatic** in the tools.
2. Select **Trimmomatic flexible read trimming tool for Illumina NGS data**.
3. Single-end or paired-end reads? > **Paired-end (as collection)**.
4. Average quality required = **30**.
5. Insert Trimmomatic Operation:
 - Select Trimmomatic operation to perform: **MINLEN**.
 - Minimum length of reads to be kept = **50**.
6. Select **Run tool**.

Galaxy Europe

Herramientas

trimmomatic

Trimmomatic flexible read trimming tool for Illumina NGS data

WORKFLOWS

Todos los flujos de trabajo

Tool Parameters

Single-end or paired-end reads?

Paired-end (two separate input files)

Input FASTQ file (R1/first of pair)

1: ERR5319322_1.fastq.gz

Input FASTQ file (R2/second of pair)

2: ERR5319322_2.fastq.gz

Perform initial ILLUMINACLIP step?

no

Cut adapter and other Illumina-specific sequences from the read

Trimmomatic Operation

1: Trimmomatic Operation

Select Trimmomatic operation to perform

Sliding window trimming (SLIDINGWINDOW)

Number of bases to average across

4

Average quality required

39

+ Insert Trimmomatic Operation

Quality score encoding (optional)

Nothing selected

The phred+64 encoding works the same as the phred+33 encoding, except you add 64 to the phred score to determine the ascii code of the quality character. You will only find phred+64 encoding on older data, which was sequenced several years ago. FASTQC can be used in order to identify the encoding type.

Output trimlog file?

☐ No (-trimlog)

Output trimmomatic log messages?

☐ No

these are the messages written to stderr (eg. for use in MultiQC)

History

search datasets

Illumina preprocessing

71.6 MB

14: fastp on data 2 and data 1: JSON report

13: fastp on data 2 and data 1: HTML report

12: fastp on data 2 and data 1: Read 2 output

11: fastp on data 2 and data 1: Read 1 output

10: FastQC on data 2: RawData

9: FastQC on data 2: Webpage

8: FastQC on data 1: RawData

7: FastQC on data 1: Webpage

2: ERR5319322_2.fastq.gz

1: ERR5319322_1.fastq.gz

Galaxy Europe

Herramientas

trimmomatic

Trimmomatic flexible read trimming tool for Illumina NGS data

WORKFLOWS

Todos los flujos de trabajo

Tool Parameters

2: Trimmomatic Operation

Select Trimmomatic operation to perform

Drop reads below a specified length (MINLEN)

Minimum length of reads to be kept

50

+ Insert Trimmomatic Operation

Quality score encoding (optional)

Nothing selected

The phred+64 encoding works the same as the phred+33 encoding, except you add 64 to the phred score to determine the ascii code of the quality character. You will only find phred+64 encoding on older data, which was sequenced several years ago. FASTQC can be used in order to identify the encoding type.

Output trimlog file?

☐ No (-trimlog)

Output trimmomatic log messages?

☐ No

these are the messages written to stderr (eg. for use in MultiQC)

Additional Options

Email notification

☐ No

Send an email notification when the job completes.

Run Tool

Help

What it does

Trimmomatic performs a variety of useful trimming tasks for Illumina paired-end and single ended data.

This tool allows the following trimming steps to be performed:

- ILLUMINACLIP:** Cut adapter and other Illumina-specific sequences from the read
 - If **Always keep both reads (PE specific/palindrome mode)** is True, the reverse read will also be retained in palindrome mode. After read-through has been detected by palindrome mode, and the adapter sequence removed, the reverse read contains the same sequence information as the forward read, albeit in reverse complement. For this reason, the default behaviour is to entirely drop the reverse read. Retaining the reverse read may be useful e.g. if the downstream tools cannot handle a combination of paired and unpaired reads.
- SLIDINGWINDOW:** Perform a sliding window trimming, cutting once the average quality within the window falls below a threshold
- MINLEN:** Drop the read if it is below a specified length
- LEADING:** Cut bases off the start of a read, if below a threshold quality
- TRAILING:** Cut bases off the end of a read, if below a threshold quality
- CROP:** Cut the read to a specified length
- CRAP:** Cut the specified number of bases from the start of the read

History

search datasets

Illumina preprocessing

71.6 MB

14: fastp on data 2 and data 1: JSON report

13: fastp on data 2 and data 1: HTML report

12: fastp on data 2 and data 1: Read 2 output

11: fastp on data 2 and data 1: Read 1 output

10: FastQC on data 2: RawData

9: FastQC on data 2: Webpage

8: FastQC on data 1: RawData

7: FastQC on data 1: Webpage

2: ERR5319322_2.fastq.gz

1: ERR5319322_1.fastq.gz

Trimmomatic **does not** perform statistics over trimmed reads, so we need to perform FastQC again over the Trimmomatic results.

- Try to do it on your own.
- How can I improve the quality of my data?

This hands-on history URL: <https://usegalaxy.eu/u/svarona/h/illumina-preprocessing>

2. Nanopore Quality control and trimming

Title	Galaxy
Training dataset:	The data we are going to manage now corresponds to Nanopore amplicon sequencing data using ARTIC network primers for the SARS-CoV-2 genome. From the Fast5 files generated by the ONT software, we are going to select the pass reads, so they are already filtered by quality.
Questions:	• How do I know if my Nanopore data was correctly sequenced?
Objectives:	• Perform a quality control in raw Illumina reads. • Perform a quality trimming in raw Nanopore reads. • Perform a quality control in trimmed Nanopore reads.
Estimated time:	15 min

2.1. Quality control

To run the quality control over the samples, follow these steps:

1. [Create a new history has explained yesterday](#) named **Nanopore quality**.
2. [Upload data as seen yesterday](#), copy and paste the following URLs:

```
https://raw.githubusercontent.com/nf-core/test-datasets/viralrecon/nanopore/minion/fastq_pass/barcode01/FA093606_pass_barcode01_7650855b_0.fastq
https://raw.githubusercontent.com/nf-core/test-datasets/viralrecon/nanopore/minion/fastq_pass/barcode01/FA093606_pass_barcode01_7650855b_1.fastq
https://raw.githubusercontent.com/nf-core/test-datasets/viralrecon/nanopore/minion/fastq_pass/barcode01/FA093606_pass_barcode01_7650855b_2.fastq
```

2.1.1. PycoQC

To use PycoQC we need to use the [sequencing_summary.txt](#) provided by the Nanopore sequencing machine.

[Upload data as seen yesterday](#), copy and paste the following URL:

```
https://raw.githubusercontent.com/nf-core/test-datasets/viralrecon/nanopore/minion/sequencing_summary.txt
```

1. Search for the **Pycoqc** tool.
2. Select **Pycoqc quality control for Nanopore sequencing data**.

- Herramientas

pycoqc

Cargar Datos

Show Sections

Pycoqc

FLUJOS DE TRABAJO

Todos los flujos de trabajo

Flujo de Trabajo

Visualizar

Datos Compartidos

Ayuda

Usuario

Pycoqc (Galaxy Version 2.5.2+galaxy0)

A sequencing_summary file

16: sequencing_summary.txt

Must be generated by Albacore 1.0.0 + (read_fast5_basecaller.py) / Guppy 2.1.3+ (guppy_basecaller) (--summary_file)

An alignment file generated by aligning the reads described in the sequencing_summary file

No bam dataset available.

(--bam_file)

Output JSON Summary File?

Yes

Minimum quality to consider a read as 'pass'

7

(--min_pass_qual)

4 Execute

General run summary

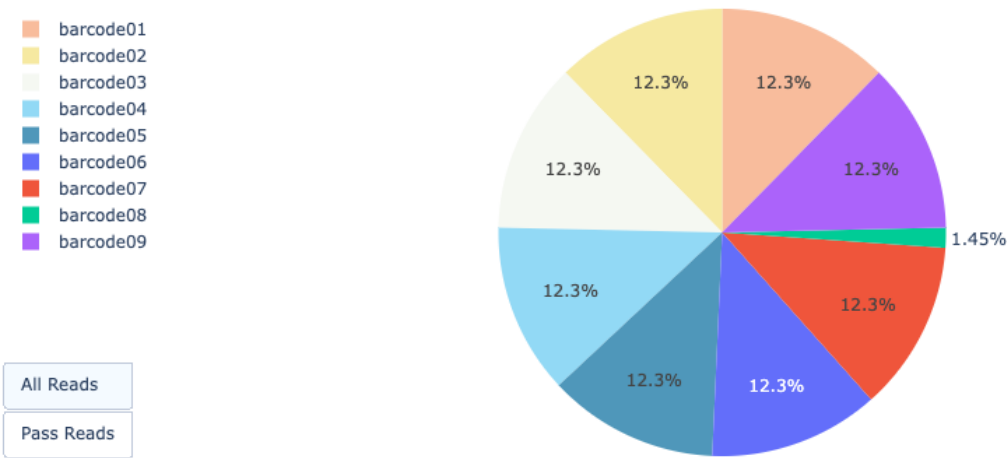
Status	Run Duration (h)	Active Channels	Number of Runids	Number of Barcodes
All Reads	7.98	496	1	9
Pass Reads	7.98	494	1	9

Basecall summary

Status	Reads	Bases	N50	Median Read Length	Median PHRED score
All Reads	2.449000e+4	1.284332e+7	515	514	13.055
Pass Reads	2.435200e+4	1.276470e+7	515	514	13.067

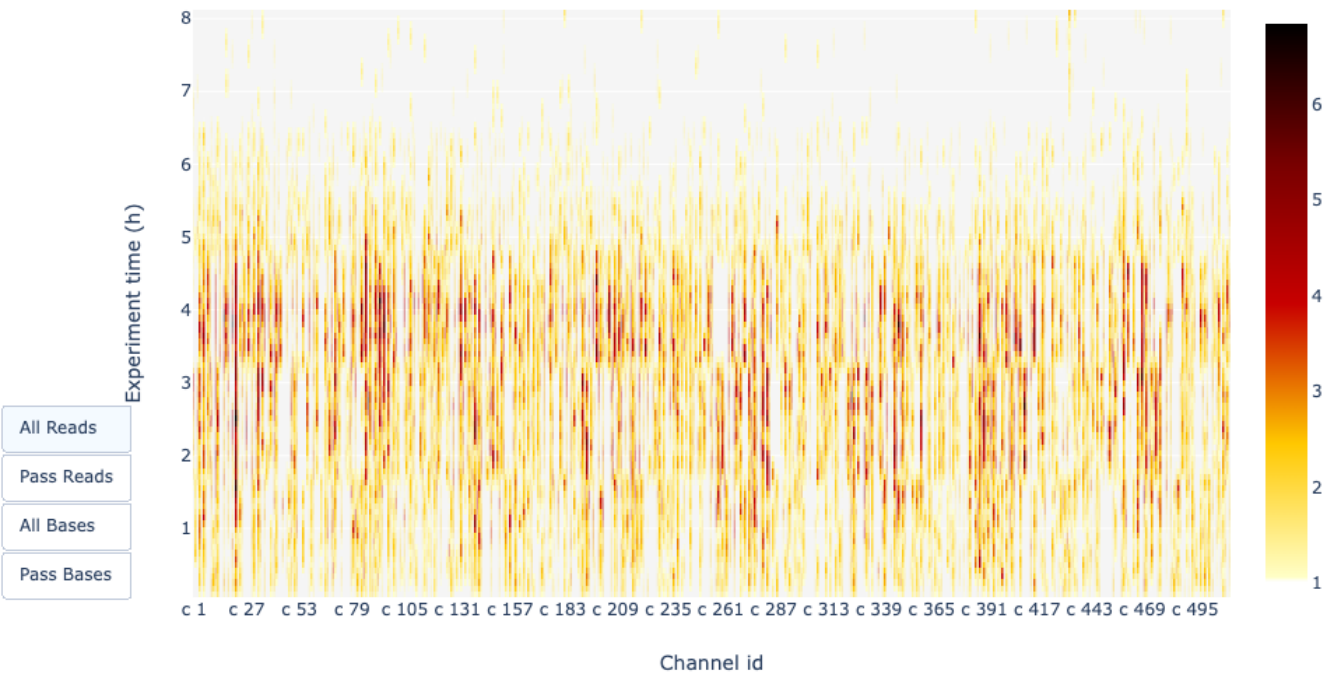
- ▶ How many reads do the samples have?
- ▶ Do you understand all the plots?

Number of reads per barcode:



This plot shows the number of reads per barcode, which means the number of reads per sample to be demultiplexed. In a good experiment, all the barcodes should have the same number of reads. In this training, we only use reads from the barcode01 sample, but we can see that barcode08 couldn't be correctly sequenced.

Channel activity over time:



It gives an overview of available pores, pore usage during the experiment, inactive pores and shows if the loading of the flow cell is good (almost all pores are used). In this case, the vast majority of channels/pores are inactive (white) after the 6h of experiment, so the run should have been finished at that time. You would hope for a plot that it is dark near the X-axis, and with higher Y-values (increasing time) doesn't get too light/white. Depending on whether you choose "Reads" or "Bases" on the left, the color indicates either the number of bases or reads per time interval.

► How do I check whether my Nanopore data was correctly sequenced?

2.2. Trimming

When Nanopore reads are being sequenced, the **MinKnown** software splits Fast5 reads into quality **pass** and quality **fail**. As we will select only Fast5 **pass reads**, we won't need to perform a quality trimming, so even if we see that the reads have a bad Phred score, we know that the ONT software considered the reads as "**good quality**".

Then we will only be performing a **read length trimming**. Since we are using amplicon sequencing data, we won't be expecting reads smaller than 400 nucleotides, nor higher than 600, which would obviously correspond to **chimeric reads**.

2.2.1. ARTIC

1. Search for **artic** tool.
2. Select **ARTIC guppyplex Filter Nanopore reads by read length and (optionally) quality**.
3. Structure of your input data: **Multiple input datasets per sample**.
4. While pressing the *Ctrl* key, **select the three samples**.
5. Remove reads longer than = **600**.
6. Remove reads shorter than = **300**.
7. Do not filter on quality score (speeds up processing) = **Yes (we had already selected pass reads)**.
8. Run Tool.

Galaxy Europe

Workflow Visualize Shared Data Help User

Tools

artic 1

Upload Data

Show Sections

ARTIC minion Build consensus sequence and call variants from amplicon-based nanopore sequence data

ARTIC guppyplex Filter Nanopore reads by read length and (optionally) quality 2

Analyze particles of binary image

WORKFLOWS

All workflows

ARTIC guppyplex Filter Nanopore reads by read length and (optionally) quality (Galaxy Version 1.2.1+galaxy2)

Structure of your input data 3

Multiple input datasets per sample

Partial sequencing datasets for your sample

3: FAO93606_pass_barcode01_7650855b_2.fastq
2: FAO93606_pass_barcode01_7650855b_1.fastq
1: FAO93606_pass_barcode01_7650855b_0.fastq 4

Multiple datasets selected here will get combined into a single output for a single assumed sample. Select a nested list to have its inner lists interpreted as data from one sample each and to obtain one output per inner list.

Remove reads longer than 5

600

remove reads greater than this number of base pairs

Remove reads shorter than 6

300

remove reads less than this number of base pairs

Eliminate reads with a mean base quality score of less than

7

Set to 0 to skip the quality check.

Email notification

No

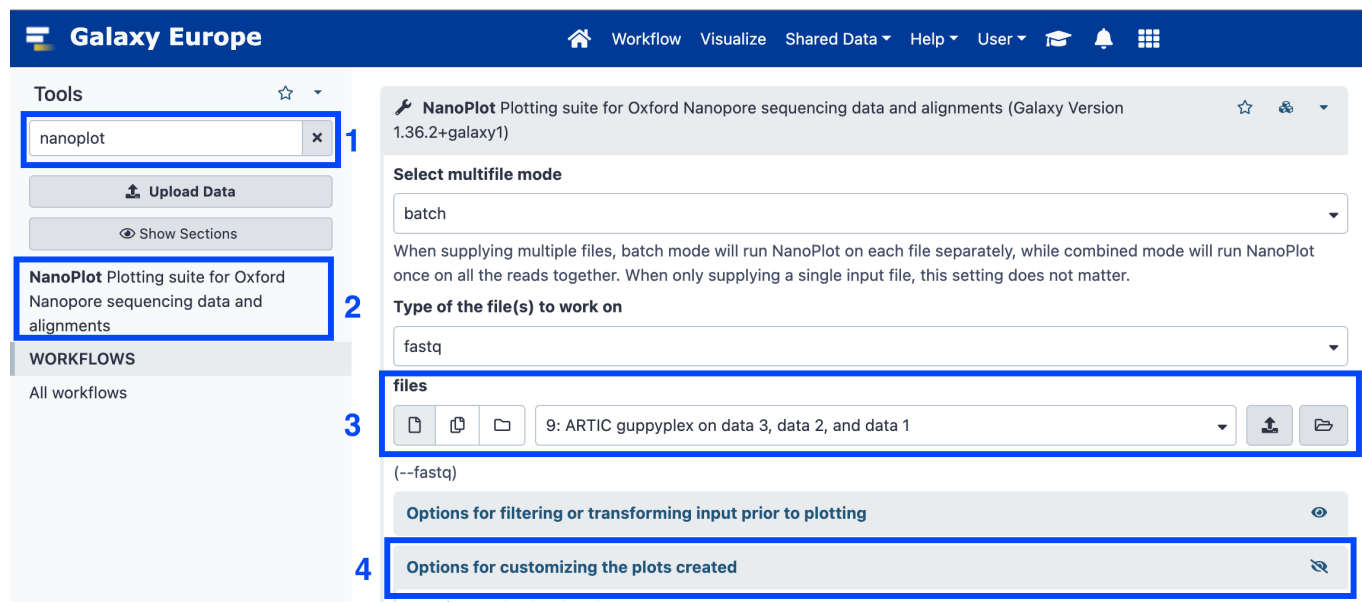
Send an email notification when the job completes.

✓ Execute 7

2.2.2. Nanoplot

Now we are going to run **NanoPlot** on filtered data:

1. Search for the **NanoPlot** tool and select **NanoPlot Plotting suite for Oxford Nanopore sequencing data and alignments**
2. Run the tool as follows:
 - In the *files* part, select **ARTIC output file**.
 - Display **Options for customizing the plots created**:
 - **Specify the bivariate format of the plots** > *Select all*
 - **Show the N50 mark in the read length histogram** > *Yes*
3. Select **Run Tool**.



Galaxy Europe

Workflow Visualize Shared Data Help User

Tools

nanoplot

Upload Data

Show Sections

NanoPlot Plotting suite for Oxford Nanopore sequencing data and alignments

WORKFLOWS

All workflows

NanoPlot Plotting suite for Oxford Nanopore sequencing data and alignments (Galaxy Version 1.36.2+galaxy1)

Select multifile mode

batch

When supplying multiple files, batch mode will run NanoPlot on each file separately, while combined mode will run NanoPlot once on all the reads together. When only supplying a single input file, this setting does not matter.

Type of the file(s) to work on

fastq

files

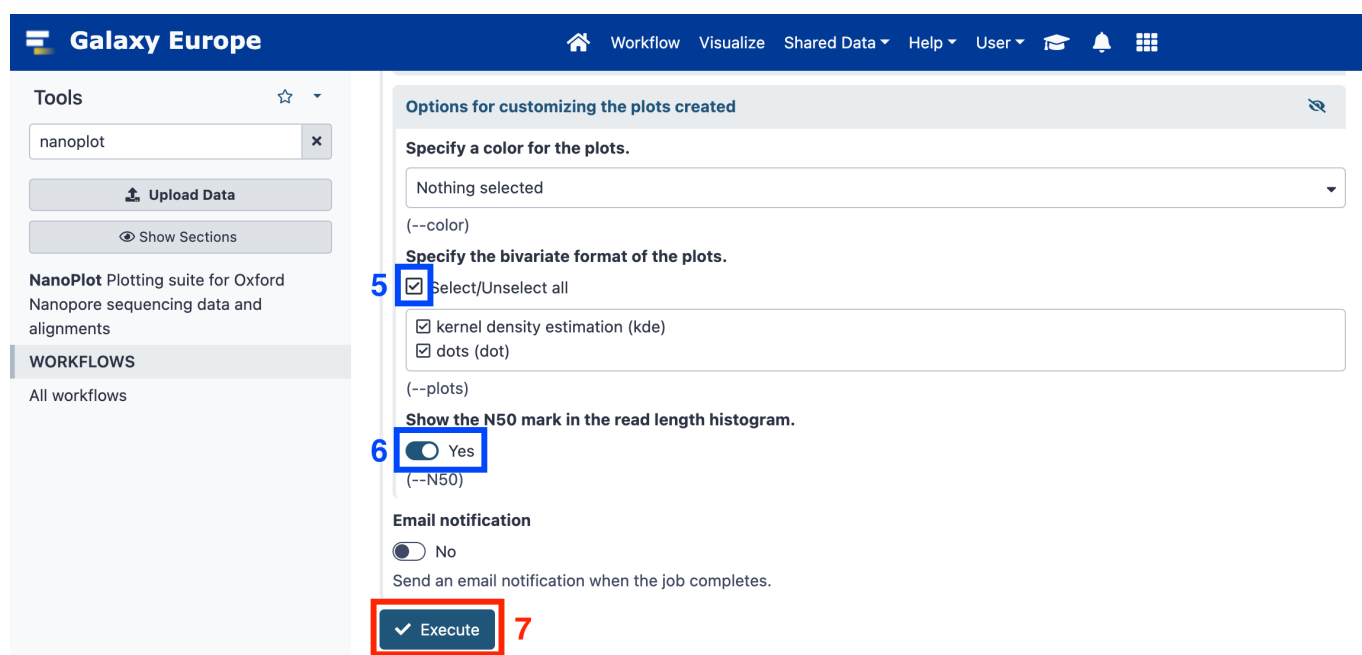
9: ARTIC guppyplex on data 3, data 2, and data 1

(--fastq)

Options for filtering or transforming input prior to plotting

Options for customizing the plots created

Specify a color for the plots.



Galaxy Europe

Workflow Visualize Shared Data Help User

Tools

nanoplot

Upload Data

Show Sections

NanoPlot Plotting suite for Oxford Nanopore sequencing data and alignments

WORKFLOWS

All workflows

Options for customizing the plots created

Specify a color for the plots.

Nothing selected

(--color)

Specify the bivariate format of the plots.

☒ Select/Unselect all

☒ kernel density estimation (kde)

☒ dots (dot)

(--plots)

Show the N50 mark in the read length histogram.

☒ Yes

(--N50)

Email notification

☐ No

Send an email notification when the job completes.

Execute

Questions

- ▶ Did our data length and quality improve?
- ▶ How many reads did we lose during the trimming step?

This hands-on history URL: <https://usegalaxy.eu/u/svarona/h/nanopore-quality>.